

Introduction to Engineering Labs





About the Course

Note that labs are an essential part of science in which students engage with the Things they are reading about and practice the scientific method.

This topic is included in the following course(s): Science: Introduction to Engineering



Scheduling

GRADE	SCHEDULE INFO.	BOOKS
9-12	Introduction to Engineering Labs 1 time/week 60 min	

Sample Weekly View

Day 1	Day 2	Day 3	Day 4	Day 5
Science: Introduction to Engineering				
Introduction to Engineering Lessons Nature Walks & Scouting: Grades 9-12	Introduction to Engineering Lessons	Introduction to Engineering Lessons	Introduction to Engineering Lessons Nature Notebook: Grades 9-12	Introduction to Engineering Lessons Introduction to Engineering Labs



Planning & Prep

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LINKS: Click text or scan the QR code in the top corner of the lesson plan pages to view online resources associated with the lessons.

Responsibility for previewing all links rests with the teacher. All links were checked at the time of publication; however, websites change frequently and may contain objectionable content. Please report broken links by contacting us through our website.



Supplies

For supply list details and basic supplies helpful to have on hand, click the links or scan the QR code below.

∞ [View Basic Supplies](#)

(No Subject Supplies Assigned)



Quick Links

(No Quick Links Assigned)

Click THIS text
or scan the QR
code for links.



Introduction to Engineering Labs

How To Approach Labs



Introduce

- Regardless of how many days are required to complete a particular activity, every Science Lab has the same flow, which follows the scientific method.
- Relevant concept(s) are introduced in the text.
- Your notebook entry begins with the introductory/prelab narration, including relevant information that you have read or previously experienced, what you plan to do in the lab, and any hypothesis or anticipated result.



Lab Procedure

- Perform the procedure according to the instruction, recording in your notebook what you do as it happens. This can be a challenge, but is an extremely important skill.
- Record all data and observations in the lab notebook.



Analysis & Conclusions

- After all data is collected, analyze the results by considering how the data reflects the introduced concepts and whether the hypothesis is supported by the data.
- If the data and observations do not support the hypothesis, reflect on why and what further testing would be interesting or helpful.

Introduction to Engineering Labs

[Click THIS text or scan the QR code for links.](#)



Term 1

WEEK 1 60m Introduction to Engineering Labs - Lesson 1

Materials: internet access

→ PROJECT

If there is a design problem in your home or community that you would like to tackle, use this time to research, design, and execute your solution.

If you'd like to sample different areas of engineering, begin with the structural engineering project linked, The Leaning Tower of Pasta. Or you might find your own idea by exploring the Science Buddies website. The filters on the left side of the page allow you to narrow your results. Once you have chosen a project, let your teacher know if you need any materials. Your project might take the whole year or you might finish it in just a few weeks. Whenever you finish your project, select another one! Record your work and your observations in your lab notebook.

∞ Website Link: The Leaning Tower of Pasta

∞ Website Link: Science Buddies

WEEK 2 60m Introduction to Engineering Labs - Lesson 2

Materials: by student choice & notebook

→ PROJECT

Continue working on your project.

WEEK 3 60m Introduction to Engineering Labs - Lesson 3

Materials: by student choice & notebook

→ PROJECT

Continue working on your project.